



Design Guide for Saudi Arabia

Minimum design standards for public health services



Design Guide for the Kingdom of Saudi Arabia:

Minimum design standards for public health services

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Note from the publisher

This publication is primarily intended to provide guidance to those responsible for the design, installation, commissioning, operation and maintenance of building services. It is not intended to be exhaustive or definitive and it will be necessary for users of guidance to exercise their own professional judgement when deciding whether to abide by or depart from it.

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This document aims at establishing the minimum design standards for the design of public health systems for buildings in the Kingdom of Saudi Arabia. It does not relieve the designer of responsibility for accurately determining design arrangements and/or complying with the local authority requirements and/or relevant international codes or regulatory requirements. It should be used for initial design guidance, and provides minimum design requirements for specific design issues.

This guide has been produced in good faith by the CIBSE UAE Technical Committee to promote best engineering practices in the Kingdom of Saudi Arabia, without legal or financial liability of CIBSE UAE or the authors, in order to assist engineers coming from elsewhere in the world to work in the Kingdom of Saudi Arabia. It should be read alongside existing CIBSE guidance relevant to the region, including *Buildings for Extreme Environments: Arid*.

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1 Design codes, standards and guidance

The design, specification and installation of public health services should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and regulations applicable in the Kingdom of Saudi Arabia (KSA):

- Saudi Building Codes (SBC)
- Saudi Civil Defence Requirements (SCD)
- Saudi Arabia National Water Company (NWC)
- MEWA Water and Wastewater Master Plan Guidelines
- Saudi Civil Defence Requirements (SCD)
- International Plumbing Code (IPC) (ICC, 2021a)
- American Society of Plumbing Engineers (ASPE)
- Uniform Plumbing Code (UPC) (IAPMO, 2021)
- British Standards (BS)
- European Norms (EN)
- International Standards (ISO)
- Water Research Centre (UK) publications
- Chartered Institute of Plumbing and Heating Engineering (CIPHE) (formerly the Institute of Plumbing (IoP)) good practice and design guides.

The public health systems covered in this section are as follows:

- local green building requirements
- domestic water systems (hot and cold)
- drainage systems
- rainwater systems
- gas systems.

The main objectives should be to:

- ensure that all water supplies are clear, odourless, tasteless and wholesome for use
- ensure that the system is protected against Legionella
- provide potable water in the amount and at the pressure required by the building occupancy and type of plumbing fixtures utilised in the building
- prevent contamination from non-potable liquids, solids or gases
- prevent water hammer conditions by providing air chambers and correct bracketry
- minimise leakage and wastage
- make use of sustainable material and methods
- minimise the consumption of energy.

2 Local green building requirements

Green building requirement shall comply with SBC-1001: *Saudi Green Construction Code* (SBCNC, 2018a) and Mostadam rating systems:

- Mostadam Rating System: Communities D+C Manual (MoH, 2019a)
- Mostadam Rating System: Residential Buildings O+E Manual (MoH, 2019b)

Table 2.1 below defines the water services maximum flow rates based on Mostadam *O+E Manual* (MoH, 2019b), Table W-01.2, 'Maximum flow rate or volume per water fixture'.

All water supply connections to appliances should be fitted with an automatic flow limiting valve (AFL) to offer further reductions to the overall water consumption of individual appliances. The number of credit points achieved is dependent on the percentage reduction of indoor water consumption compared to the baseline (Table W-01.1 (MoH, 2019b)).

Table 2.1 Maximum flow rate or volume for each water fixture and appliance (source: Mostadam *O+E Manual* (MoH, 2019b), Table W-01.2)

Appliance	Maximum flow rate
Lavatory faucet	5.7 litre/min. at 414 kPa
Kitchen faucet	6.84 litre/min. at 414 kPa
Urinal	1.89 litres per flush
Water closet: tank type	4.86 litres per flush (Watersense labelled)
Water closet: flushometer type	4.86 litres per flush
Shower head/hand-held spray	7.6 litre/min. at 552 kPa (Watersense labelled)
Drinking fountain (manual)	2.66 litre/min.
Washing machine	125.9 litres per cycle
Dishwasher	24 litres per cycle

3 Domestic cold water services, design criteria, codes, standards and regulations

The design, specification and installation of cold water systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and regulations applicable to KSA:

- SBC-701: *Saudi Plumbing Code* (SBCNC, 2018b)
- *International Plumbing Code* (IPC) (ICC, 2021a)
- *ASPE Plumbing Engineer Design Handbook*: Volume 1-2017, Volume 2-2018, Volume 3-2019 and Volume 4-2016 (ASPE, 2016–2018)
- CIBSE Guide G: *Public health and plumbing engineering* (CIBSE, 2014)
- BS EN 806* (BSI, 2000–2010)
- BS 8558* (BSI, 2015)
- HSE L8: *Legionnaires' disease. The control of legionella bacteria in water systems* (HSE, 2013)
- HSE HSG274: *Legionnaires' disease. Technical guidance* (three Parts) (HSE, 2014)
- *Guidelines for drinking-water quality* (WHO, 2017)
- *Plumbing Engineering Services Design Guide* (IoP, 2002).

Maintain a maximum velocity in the water supply piping as per Table 3.1 below to prevent noise and decrease the danger of surge pressure shock.

* Code to be followed only when local and US codes do not provide any specific information)

Table 3.1 Maximum velocity in water supply piping to prevent noise and decrease the danger of surge pressure shock (source: *Plumbing Engineering Services Design Guide* (IoP, 2002), Table 17)

Location	Noise rating (NR)	Metal pipes (m/s)	Plastic pipes (m/s)
Service duct, riser, shaft, plant room	50	2.0	2.5
Service enclosure, ceiling void	40	1.5	1.5
Circulation area, entrance corridor	35	1.5	1.5
Seating area, lecture/meeting room	30	1.25	1.25
Bedroom	25	1.0	1.0
Theatre, cinema	20	0.75	0.75
Recording studios	Less than 20	0.5	0.5

4 Water storage design criteria

The following storage figures have been taken from Ministry of Environment, Water & Agriculture's Master Plan Guidelines (MEWA, 2018) to calculate the one-day total storage requirement.

A total of one to two days' storage should be allowed for in all buildings, therefore two times the daily volume should form the basis of the total on-site water storage.

The water supply demand and storage requirements should be validated or approved by MEWA in the early project stage.

Table 4.1 Water storage requirements (MEWA, 2018)

Type	Storage (litre/person per day)
Residential	250–350
Mixed use residential	200–250
Villa	250–300
Labour accommodation	80–150
Commercial and retail	60–100
Mixed use commercial	60–100
Restaurant (per meal)	10–15
Hotel (per room including workers and visitors)	200–300
Workshop	60
Office	60–80
School	60–80
Mosque	10–60
Public amenity	10–60
Entertainment and leisure	60
Theatre	10–60
University	60–80
Manufacturing	60–80
Logistics and business zone	60–80
Nursery	60–80
Events venue	10–60
Gym/swimming pool	100

Note: The water demand should also be validated by the developer and might deviate from the MEWA water demand requirement.

5 Pipe sizing

The size of water service mains, branch mains and risers shall be determined accordingly to water supply demand (litre/minute), available water pressure (kPa) and friction loss due to the water meter and developed length of pipe (m), including equivalent length of fittings.

Supply load in the building water distribution system shall be determined by total load on the piped being sized, in terms of water supply fixture units (WSFU), as shown in SBC-701 (SBCNC, 2018b), Table E103.3(2).

Table 5.1 Water supply fixture units (WSFU) for water fittings

Fixture	Occupancy	Type or supply control	Load values, in water supply fixture units (WSFU)		
			Cold	Hot	Total
Bathroom group	Private	Flush tank	2.7	1.5	3.6
Bathroom group	Private	Flushometer valve	6.0	3.0	8.0
Bidet	Private	Faucet	1.5	1.5	2.0
Combination fixture	Private	Faucet	2.25	2.25	3.0
Dishwashing machine	Private	Automatic	—	1.4	1.4
Drinking fountain	Offices, etc.	20 mm valve	0.25	—	0.25
Kitchen sink	Private	Faucet	1.0	1.0	1.4
Laundry trays (1 to 3)	Private	Faucet	1.0	1.0	1.4
Lavatory	Private	Faucet	0.5	0.5	0.7
Lavatory	Public	Faucet	1.5	1.5	2.0
Service sink	Offices, etc.	Faucet	2.25	2.25	3.0
Shower head	Public	Mixing valve	3.0	3.0	4.0
Shower head	Private	Mixing valve	1.0	1.0	1.4
Urinal	Public	25 mm flushometer valve	10.0	—	10.0
Urinal	Public	20 mm flushometer valve	5.0	—	5.0
Urinal	Public	Flush tank	3.0	—	3.0
Water closet	Private	Flushometer valve	6.0	—	6.0
Water closet	Private	Flush tank	2.2	—	2.2
Water closet	Public	Flushometer valve	10.0	—	10.0
Water closet	Public	Flush tank	5.0	—	5.0
Water closet	Public or private	Flushometer tank	2.0	—	2.0

Note: To convert the WSFU into flow rates, the corresponding value should be read from SBC-701, Table E103.3(3)

6 Legionella control

Legionella control should conform to, but not be limited to, the following standards, codes of practice and regulations applicable to KSA:

- HSE L8: *Legionnaires' disease. The control of legionella bacteria in water systems* (HSE, 2013)
- HSE HSG274: *Legionnaires' disease. Technical guidance* (three Parts) (HSE, 2014)
- *Guidelines for drinking-water quality* (WHO, 2017)

Bacterial protection should be incorporated to protect against the risk of *Legionella*; the acceptable primary forms of water treatment are as below:

- Temperature control: It is recommended that hot water should be stored at 60 °C and distributed so that, after one minute of running, a temperature of at least 50 °C is attainable at outlets. Cold water storage and distribution should be at 20 °C or below.
- Pasteurisation: Water systems must be flushed at least every six month with water of not less than 70 °C for 5 minutes (or an equivalent temperature/time combination) or treated with chlorine to provide minimum free chlorine of 1–2 mg/l at all outlets.
- Cu-Ag ionization treatment.
- Chlorination: This is the process of applying any of the chlorine-containing compounds such as sodium hypochlorite (bleach solution) to the water to achieve the necessary destruction of all bacteria. The maximum chlorine dosing shall not exceed 0.5 mg/l and the residual chlorine at any fixtures within the water distribution system shall be at least 0.2 ppm (mg/l) as per WHO standards.
- Chlorine dioxide: This is an oxidizing biocide capable of reacting with a wide range of organic substances. There are commercial systems available that release chlorine dioxide from a stabilized precursor solution into water systems. The maximum value of chlorine dioxide in water should not exceed 0.5 mg/l as chlorine dioxide. Chlorine dioxide level of 0.5 mg/l can if properly managed, be effective against planktonic *Legionella* in both hot and cold water systems.
- Ultraviolet light (UV), and ozone (O₃): The previous treatment systems are effective throughout the water system downstream to the point of application. UV and O₃ can be used to treat water at locations very close to the point of application.

The chlorination system should be designed carefully to ensure that there is no over-chlorination dosing to the system, as this could affect the life span of the pipe material.

7 Domestic hot water services, design criteria, codes, standards and regulations

The design, specification and installation of hot water systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and specifications:

- SBC-701: *Saudi Plumbing Code* (SBCNC, 2018b)
- CIBSE Guide G: *Public health and plumbing engineering* (CIBSE, 2014)
- BS EN 806: *Specifications for installations inside buildings conveying water for human consumption. Operation and maintenance* (five Parts) (BSI, 2000–2010)
- BS 8558: *Guide to the design, installation, testing and maintenance of services supplying water for domestic use within buildings and their curtilages. Complementary guidance to BS EN 806.* (BSI, 2015)
- Chartered Institute of Plumbing and Heating Engineers (CIPHE) guidance
- American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE) guidance
- *Plumbing Engineering Services Design Guide* (IoP, 2002)
- *ASPE Plumbing Engineering Design Handbook*: Volume 1-2017, Volume 2-2018, Volume 3-2019 and Volume 4-2016 (ASPE, 2016–2018)
- HSE L8: *Legionnaires' disease. The control of legionella bacteria in water systems* (HSE, 2013)
- BRE IP14/03: *Preventing hot water scalding in bathrooms using TMVs* (2003)

The following are design temperatures considered for the domestic hot water (DHW) system:

- Minimum DHWS storage temperature: 60 °C
- Minimum temperature at outlet: 55 °C
- Minimum temperature of return to storage cylinders: 50 °C

The hot water temperature for appliances, where applicable, should be temperature-controlled via a thermostatic mixing valve (TMV) as detailed within the Table 7.1.

Table 7.1 Recommended temperatures for thermostatic mixing valves
(source: *Plumbing Engineering Services Design Guide* (IoP, 2002))

Appliance	Application	Thermostatic mixing valve	
		Temp. (°C)	Type
Bidet	All	38	3
Shower	All	41	3
Wash hand basin	Public	41	1
Wash hand basin	Private	43	3
bath	All	44	3

7.1 Centralised hot water systems

The hot water storage figures given in Table 7.2 have been taken from Table 6 of the *Plumbing Engineering Services Design Guide* (IoP, 2002) assist in calculating daily hot water requirements and storage for a building, based on a two-hour reheat period, an inlet temperature of 10 °C and a stored temperature of 65 °C. The design engineer should make the necessary adjustments depending on the actual reheat period, inlet water temperature and storage temperature.

Table 7.2 Recommended hot water storage temperatures (source: *Plumbing Engineering Services Design Guide* (IoP, 2002))

Type of building	Daily consumption (litres)	Stored (litres)	Unit
Dwellings:			
— 1 bedroom	115	115	Per bedroom
— 2 bedrooms	75	115	Per bedroom
— 3+ bedrooms	55	115	Per bedroom
— student, en-suite	70	20	Per bedroom
— student, communal	70	20	Per bed space
— nurses home	70	20	Per bed space
— children's home	70	25	Per bed space
— elderly sheltered	70	25	Per bed space
— elderly care home	90	25	Per bed space
Hotels:			
— budget	115	35	Per bedroom
— Travel Inn/Lodge	115	35	Per bedroom
— luxury (4/5-star)	135	45	Per bedroom
Offices and general work places:			
— with canteen	15	5	Per person
— without canteen	10	5	Per person

Table continues

Table 7.2 Recommended hot water storage temperatures (*continued*)
(source: *Plumbing Engineering Services Design Guide* (IoP, 2002))

Type of building	Daily consumption (litres)	Stored (litres)	Unit
Shops:			
— with canteen	15	5	Per person
— without canteen	10	5	Per person
Factories:			
— with canteen	15	5	Per person
— without canteen	10	5	Per person
Schools:			
— nursery	15	5	Per pupil
— primary	15	5	Per pupil
— secondary	15	5	Per pupil
— 6th form	15	5	Per pupil
— boarding	114	25	Per pupil
Hospitals:			
— district general	200	50	Per bed
— surgical ward	110	50	Per bed
— medical ward	110	50	Per bed
— paediatric ward	125	70	Per bed
— geriatric ward	70	40	Per bed
Sports changing:			
— sports hall	20	20	Per person
— swimming pool	20	20	Per person
— field sports	35	35	Per person
— all-weather pitch	35	35	Per person
Places of assembly (excluding staff):			
— art gallery	2	1	Per person
— library	2	1	Per person
— museum	1	1	Per person
— theatre	1	1	Per person
— cinema	1	1	Per person
— bar	2	1	Per person
— night club	1	1	Per person
— restaurant	6	6	Per cover

7.2 Localised hot water systems

The hot water storage figures given in Table 7.3 below have been taken from the ASHRAE Handbook: *HVAC Applications* (ASHRAE, 2019) to assist in calculating hot water storage for localised systems.

Table 7.3 Hot water demand for fixtures (source: ASHRAE Handbook: *HVAC Applications* (ASHRAE, 2019))

Appliance	Hot water demand per fixture (litres of water per hour per fixture, calculated at a final temperature of 60 °C)									
	Apartment or house	Club	Gym.	Hospital	Hotel	Industrial plant	Office building	Private residence	School	YMCA
Basin, private lavatory	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6	7.6
Basin, public lavatory	15	23	30	23	30	45.5	23	—	57	30
Bathtub	76	76	114	76	76	—	—	76	—	114
Dishwasher	57	190–570	—	190–570	190–760	76–380	—	57	76–380	76–380
Foot basin	11	11	46	11	11	46	—	11	11	46
Kitchen sink	38	76	—	114	114	76	76	38	76	76
Laundry, stationary tub	76	106	—	106	106	—	—	76	—	106
Pantry sink	19	38	—	38	38	—	38	19	38	38
Shower	114	568	850	284	284	850	114	114	850	850
Service sink	76	76	—	76	114	76	76	57	76	76
Hydrotherapeutic bath	—	—	—	—	—	—	—	—	—	—
Hubbard bath	—	—	—	1520	—	—	—	—	—	—
Leg bath	—	—	—	2270	—	—	—	—	—	—
Arm bath	—	—	—	380	—	—	—	—	—	—
Sitz bath	—	—	—	130	—	—	—	—	—	—
Continuous flow bath	—	—	—	114	—	—	—	—	—	—
Circular wash sink	—	—	—	625	—	—	—	—	—	—
Semi-circular wash sink	—	—	—	76	76	114	76	—	114	—
Demand factor	—	—	—	38	38	57	38	—	57	—
Storage capacity factor	0.30	0.30	0.40	0.25	0.25	0.40	0.30	0.30	0.40	0.40
	1.25	0.90	1.00	0.60	0.8	1.00	2.00	0.70	1.00	1.00

8 Foul and waste water drainage systems, design criteria, codes, standards and regulations

The design, specification and installation of drainage systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and specifications:

- SBC-701: *Saudi Plumbing Code* (SBCNC, 2018b)
- SBC-702: *Saudi Private Sewage Disposal Code* (SBCNC, 2018c)
- *International Plumbing Code (IPC)* (ICC, 2021)
- *ASPE Plumbing Engineering Design Handbook: Volume 1-2017, Volume 2-2018, Volume 3-2019 and Volume 4-2016* (ASPE, 2016–2018)
- *Uniform Plumbing Code (UPC)* (IAPMO, 2021)
- BS EN 12056: *Gravity drainage systems inside buildings**
 - Part 1: *General and performance requirements* (BSI, 2000a)
 - Part 2: *Sanitary pipework, layout and calculation* (BSI, 2000b)
 - Part 3: *Roof drainage, layout and calculation* (BSI, 2000c)
 - Part 4: *Wastewater lifting plants. Layout and calculation* (BSI, 2000d)
- British Standards* (BS) and European Norms* (EN)

8.1 System flows

Peak black (foul) and grey (waste) water flows should be calculated according SBC-701 (SBCNC, 2018b). The discharge fixture unit method is the basis of design to calculate peak flow rates and pipe sizes for drainage systems.

A summary of fixtures is provided in Table 709.1 of SBC-701, reproduced below as Table 8.1.

Table 8.1 Discharge fixture units and waste sizes for various appliances (source: SBC-701 (SBCNC, 2018b))

Appliance	Discharge fixture unit (DFU)	Waste size (mm)
WC, public (6 litres per flush)	4.0	100
WC, private (6 litres per flush)	3.0	100
WC, flushometer tank (public or private)	4.0	100
Wash hand basin	1.0	32
Shower:		
— flow rate 22 l/min. or less	2	40
— flow rate >22 l/min. to 46 l/min.	3	50
— flow rate >46 l/min. to 98 l/min.	5	75
— flow rate >98 l/min. to 210 l/min.	6	100
Bath tub	2	40
Bidet	1	40
Kitchen sink	2	40
Urinal	4	40
Dishwasher	2	40
Floor gully (nominal diam. 100 mm)	2	110

* Code to be followed, only when local and US codes do not provide any specific information.

8.2 Condensate drainage, design criteria, codes, standards and regulations

Condensate waste and drain line size shall not be less than 20 mm internal diameter and shall not decrease in size from the drain pan connection to the place of condensate disposal. Where the drain pipes from more than one unit are manifold together for condensate drainage, the pipe shall be sized in accordance with Table 314.2.2 of SBC-701 (SBCNC, 2018b), reproduced below as Table 8.2.

Table 8.2 Minimum condensate pipe diameter (source: SBC-701 (SBCNC, 2018b))

Equipment capacity	Minimum condensate pipe diameter (mm)
Up to 20 tons of refrigeration	20
Up to 20 tons to 40 tons of refrigeration	25
Up to 40 tons to 90 tons of refrigeration	32
Up to 90 tons to 125 tons of refrigeration	38
Up to 125 tons to 250 tons of refrigeration	50

9 Stormwater design criteria, codes, standards and regulations

The design, specification and installation of rainwater systems should be in accordance with good engineering practice and should conform with, but not be limited to, the following standards, codes of practice and specifications:

- SBC-701: *Saudi Plumbing Code* (SBCNC, 2018b)
- *International Plumbing Code* (IPC) (ICC, 2021)
- *ASPE Plumbing Engineering Design Handbook*: Volume 1-2017, Volume 2-2018, Volume 3-2019 and Volume 4-2016 (ASPE, 2016–2018)
- MEWA Water and Wastewater Master Plan Guidelines
- BS EN 12056: *Gravity drainage systems inside buildings**
 - Part 3: *Roof drainage, layout and calculation* (BSI, 2000c)
 - Part 5: *Installation and testing, instructions for operation, maintenance and use* (BSI, 2000e)
- BS EN 752: *Drain and sewer systems outside buildings. Sewer system management*. (BSI, 2017)*

As per SBC-701, the stormwater system shall be designed based on a return period of 1:100 years with 60-minute duration.

However, as good design practice taking into consideration climate change, all rainwater downpipes should be sized based on a minimum rainfall intensity of 150 mm per hour i.e. 75 mm per hour for primary drainage and 75 mm per hour for secondary 'emergency overflow' drainage system.

* Code to be followed, only when local and US codes do not provide any specific information.

10 Fuel gas systems, design criteria, codes, standards and regulations

The design, specification and installation of fuel gas systems should be in accordance with good engineering practice and should conform to, but not be limited to, the following standards, codes of practice and regulations applicable to KSA:

- Saudi Civil Defence requirements
- NFPA 54: *National Fuel Gas Code* (NFPA, 2021) (if LPG is utilised)
- NFPA 58: *Liquefied Petroleum Gas Code* (NFPA, 2020) (if LPG is utilised)
- Institute of Gas Engineers and Managers (IGEM) Technical Standards (<https://www.igem.org.uk/technical-services/technical-gas-standards>)
- *International Fuel Gas Code* (IFGC) (ICC, 2021)

11 Water services design considerations

11.1 Cold water storage, treatment and distribution

The cold water supply design must take into account the high summer temperatures of mains cold water in KSA (peaking at approximately 42 °C at the height of summer). For some applications, such as commercial five-star hotels, hospitals, schools and sports complexes, the storage tank will need to be split to allow for a cooled cold water section. This will allow the water to be chilled to a temperature below 20 °C via plate heat exchangers served by the centralised chiller plant. The cold water supply in raw (ambient) water tanks should be filtered and treated before being stored and cooled (where required) for distribution to the building.

Additional bacterial protection should be incorporated to protect against the risk of *Legionella* as described in section 6.

11.2 Hot water generation and distribution

The primary hot water requirements for most buildings are for domestic use and for applications whereby significant volumes of hot water are required. Domestic hot water generation shall primarily be from solar energy based on a sustainability point of view.

In instances, whereby relatively low volumes are required, or in remote areas of a building or development, the use of localised hot water heating should be considered.

The type of system selected to generate the heat input to the solar thermal system should be based on high-performance solar collectors. The final systems selected will also need to be appropriate for KSA climatic conditions. Solar hot water systems should also be selected according to project size: a thermosiphon system is the better option for small villas or projects, whereas a drain-back system would be the better option for medium- to large-scale projects.

Solar hot water systems should be designed to achieve maximum efficiency. The energy back-up for the system can be water to water heat pumps, electric coil, fuel-fired boiler or any other suitable system.

11.3 Irrigation

For all new buildings, one hundred percent (100%) of the total exterior landscaping should be irrigated using non-potable water or drip or subsoil water delivery systems. Irrigation

water demand shall comply with SBC-1001: *The Saudi Green Construction Code* (SBCNC, 2018e), Section 404, 'Landscape irrigation and outdoor fountains', and Mostadam (2019 version).

All irrigation systems must incorporate, at any point that they connect to a portable water supply, backflow prevention devices which must be checked every twelve months. Testing must be in line with the manufacturer's recommended practice for field testing or any other testing regime approved by local Saudi Municipality.

11.4 Heat rejection

Potable water supplied by Saudi Municipality network must not be used for heat rejection purposes. Treated sewage effluent (TSE), sea water or recycled water should be used to meet the water demand for all heat rejections purposes. The secondary water source must be approved by local Saudi Municipality.

12 Foul and waste water drainage system design consideration

12.1 Underground drainage

There should be separate underground networks for the ground floor and upper floors. A waste water disposal system should be provided to convey the discharge from above-ground sanitation and within the building and, if applicable, any external car parking area. Foul drainage systems should be designed to meet the requirements of SBC-701 (SBCNC, 2018b) and the *International Plumbing Code* (IPC) (ICC, 2021). Reference can be made to BS EN 752 (BSI, 2017) for items not covered under SBC-701 and the IPC.

12.2 Above-ground drainage

A foul water drainage disposal system should be provided to serve the domestic sanitary fittings within new developments. The design of the system should be in accordance with of SBC-701 (SBCNC, 2018b) and the *International Plumbing Code* (IPC) (ICC, 2021). Reference can be made to BS EN 12056-2 (BSI, 2000b) for items not covered under SBC-701 and the IPC.

Foul and waste water drainage systems should be designed to safely carry away discharges from sanitary appliances in a quick and quiet manner without causing a nuisance or risk to health.

Pipework, fittings and accessories should be installed to ensure that:

- generally, all foul and waste water discharge is conveyed under gravity conditions without cross flow or back fall to prevent leakage and/or blockages occurring
- foul air from the drainage system does not enter the building
- pressure fluctuations within the pipework system do not vary by more than 50 mm water gauge
- all pipework located within main risers, shop units and plant room areas is fully accessible
- rodding access should be provided at all changes of direction, branch connections and at each floor level
- pipework systems can be adequately tested, cleaned and maintained

- pipework to be run within ceiling voids should be kept to a minimum and installed to the shortest practical route
- all pipework passing through adjoining fire compartments is fire-sleeved and sealed.

The local foul water infrastructure should be investigated to determine final connection points, agreed permissible flows and invert levels into the local authority network.

Under no circumstances should a foul water soakaway be incorporated into the design. Foul water must not be allowed to soak into the ground, as this will have an impact on groundwater conditions and could lead to public health risks to the surrounding area. Septic-type tanks should be utilised in circumstances where there is no local connectivity to the municipality sewer system. These would ideally be located external to the building in an accessible location, and need to be emptied on a regular basis by an authorised waste management company.

The foul and waste water drainage system should be designed to safely carry water from sanitary appliances via gravity discharges in a quick and quiet manner without causing a nuisance or risk to health.

The systems should comprise:

- external trapped gullies
- grease interceptors
- petrol interceptors
- lint interceptor
- sand interceptor
- oil interceptor
- neutralisation system
- interconnecting gravity drainage pipework
- manholes
- rodding access points
- linear drainage channels.

12.2.1 Grease interceptors

The foul drainage system from commercial kitchens or retail food and beverage areas should be fully protected with the use of suitably-sized grease removal units. These should be located in areas that are accessible for regular cleaning and maintenance, and should be externally located where possible. Grease separators should be designed as per SBC-701 (SBCNC, 2018b) and the *International Plumbing Code* (IPC) (ICC, 2021). Reference can be made to BS EN 1825 (BSI, 2002, 2004) for items not covered under SBC-701 and the IPC.

12.2.2 Oil interceptors

Generator rooms, plumbing workshop areas and oily waste areas should be discharged into oil interceptors before connecting to the drainage network.

12.2.3 Lint interceptor

Lint interceptors should be provided for commercial laundry prior to discharge to the drainage network.

12.2.4 Laboratory waste neutralisation system

Laboratory waste should be discharged through a neutralisation pit before connecting to the soil network, and a separate vent pipe should be provided for the laboratory system.

12.2.5 Petrol interceptors

Suitably-sized petrol interceptors should be provided for discharges from all car parking areas, and should be located in areas that are accessible for regular cleaning and maintenance. These should be provided with double-sealed recessed covers suitably rated for the expected traffic flow over manholes.

13 Storm water design consideration

A separate below-ground storm water drainage system should be provided to convey all rainwater discharge from roofs and open-ground hard-paved areas, via gravity, to the Saudi Municipality connection manhole located on the periphery of the building plot. The local surface water infrastructure should be further investigated to determine final connection points, agreed permissible flows and invert levels into the local Municipality authority network.

Flows from roofs and paved areas should be based on based on a 1:100-year storm period with a duration of 60 minutes. Rainfall intensity shall be as per specific region in KSA.

Any roof area should have at least two rainwater outlets. For large roof areas, a secondary means of overflow must be implemented.

The roofs and hard-standing areas of all buildings should be connected to the stormwater network comprising roof outlets and, where appropriate, floor gullies. Surface water should be drained from roof areas via fixed downpipes connecting into the below-ground drainage system. Downpipes should be integrated into the building fabric or mounted externally.

The rainwater drainage system should be designed to operate under gravity conditions, and should be designed to ensure that water is removed from the roof as quickly and effectively as possible to prevent any water ingress into the building. This system consists of roof outlets draining into rainwater downpipes before exiting the building to connect to the drainage infrastructure manhole.

Access points should be provided throughout the rainwater drainage system at all changes of direction and at each floor level to facilitate future maintenance.

All external hard standing areas should be drained via drainage channel and/or floor gullies, as agreed with the landscape architect, and should connect to the site-wide network. External areas should not be permitted to flood during storm conditions, drainage must be in place to deal with this event.

13.1 KSA climatic zones

The climate condition varies considerable across KSA, and therefore the design of the storm water system shall comply with local rainfall intensity and the intensity-duration-frequency (IDF) curve of the region being developed. The kingdom of Saudi Arabia is considered to be made up of three climate zones as represented by Figure 13.1 below.

The KSA storm water design guide is primarily focussed on Zone 1 and, specifically, development in Riyadh and Jeddah and proximity areas.

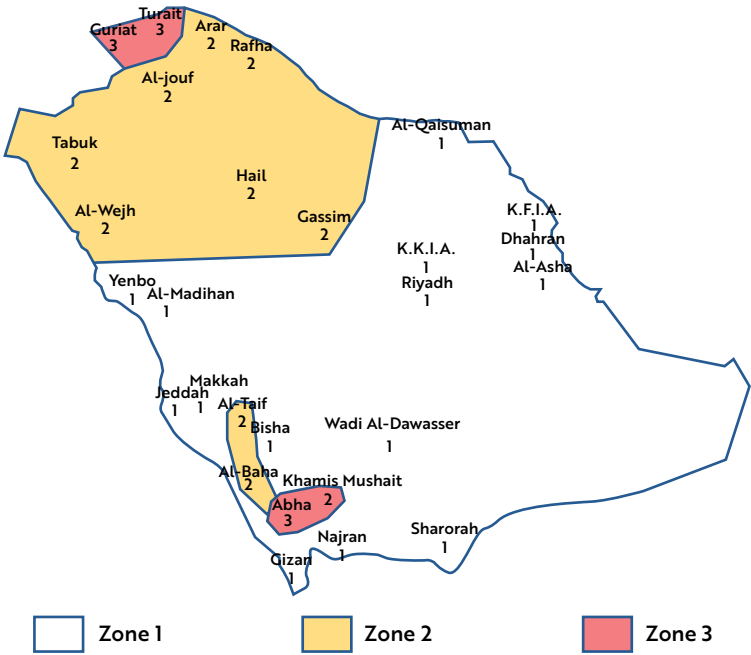


Figure 13.1 KSA climate zones (source: SBC-601, Chapter 11 (SBCNC, 2018f))

13.2 Jeddah stormwater data (Zone 1)

Figure 13.2 shows the intensity-duration-frequency (IDF) curve for Jeddah from the *Jeddah Storm Water Design Manual (JSDM)*, (AECOM, 2013).

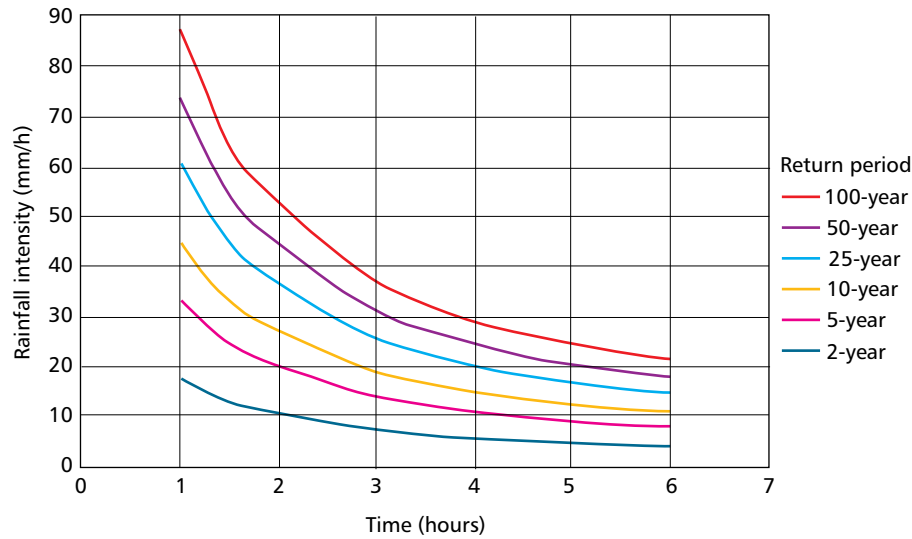


Figure 13.2 IDF curves for Jeddah

Table 13.1 Rainfall intensity for Jeddah

Return period (years)	Rainfall intensity (mm/h) for stated duration (hours)						
	1	1.5	2	3	4	5	6
2	17.6	13.0	10.6	7.3	5.6	4.6	4.0
5	33.2	24.4	20.0	14.0	10.9	9.1	7.9
10	44.6	32.9	27.0	18.8	14.8	12.3	10.7
25	60.6	44.6	36.6	25.6	20.1	16.8	14.6
50	73.5	54.0	44.3	31.0	24.4	20.4	17.8
100	87.4	64.2	52.6	36.9	29.0	24.3	21.2

13.3 Riyadh stormwater data (Zone 1)

Figure 13.3 shows the intensity-duration-frequency (IDF) curve for Riyadh from *Engineering guides for flood risk prevention criteria in Riyadh* (DAR, 2017).

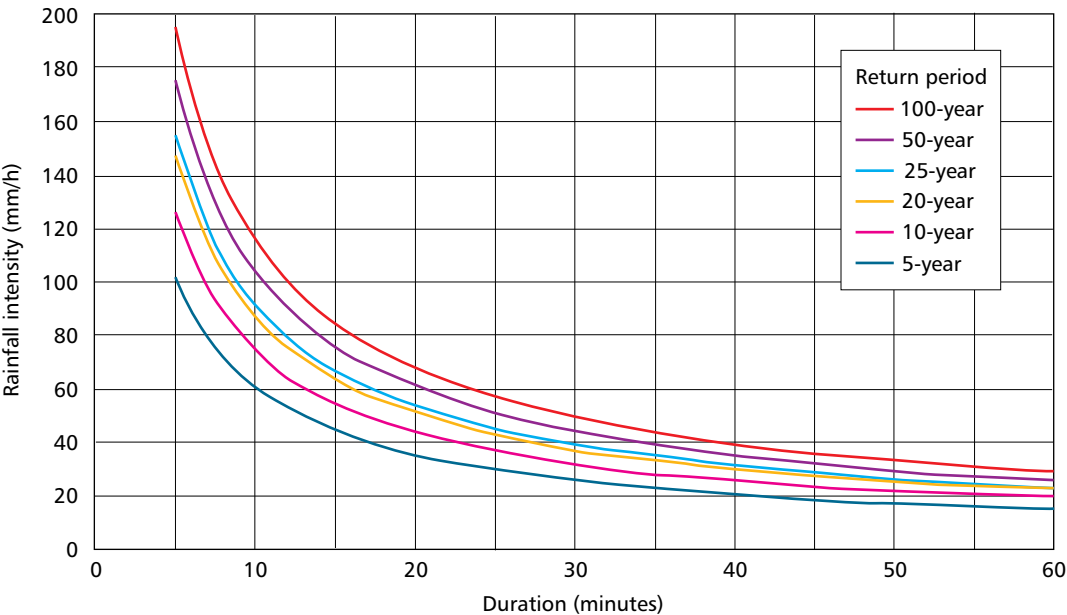


Figure 13.3 IDF curves for Riyadh

Table 13.2 Intensity-duration-frequency (IDF) for Riyadh City

Fequency (years)	Rainfall intensity (mm/h) for stated period									
	5 min	10 min	15 min	30 min	1 hr	2 hrs	3 hrs	6 hrs	12 hrs	24 hrs
5	67.2	39.1	28.9	17.1	9.92	5.94	4.28	2.57	1.50	0.85
10	103.7	60.4	44.6	26.4	15.3	9.17	6.61	3.96	2.23	1.31
15	127.1	74.1	54.7	32.4	18.8	11.2	8.10	4.86	2.84	1.61
20	148.9	86.8	64.1	37.9	22.0	13.2	9.50	5.69	3.33	1.88
25	155.8	90.7	67.0	39.6	23.0	13.8	9.93	5.95	3.48	1.97
50	176.3	102.7	75.9	44.9	26.1	15.6	11.2	6.74	3.94	2.23
100	196.3	114.4	84.5	50.0	29.0	17.4	12.5	7.50	4.38	2.48
200	215.9	125.8	92.9	54.9	31.9	19.1	13.8	8.25	4.82	2.73

13.4 Tabuk stormwater data (Zone 2)

Figure 13.4 shows the intensity-duration-frequency (IDF) curve for Tabuk based on MoMRA (Ministry of Municipal, Rural Affairs and Housing) IDF curve.

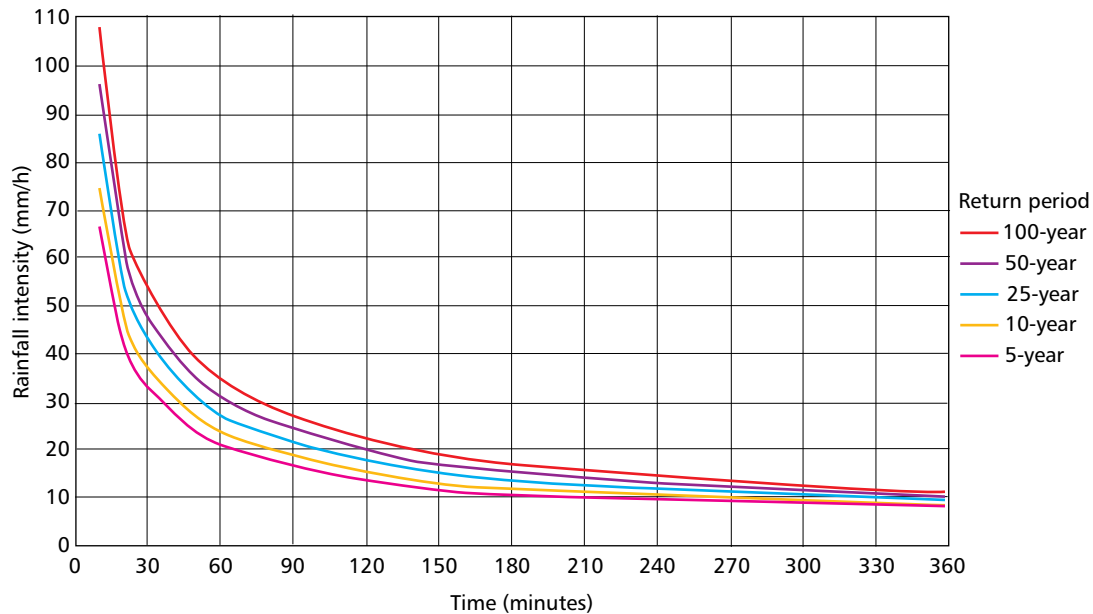


Figure 13.4 IDF curves for Tabuk

13.5 Soakaway design

Should a connection not be local to the site or development, the use of suitably-sized soakaways is permissible. These should be designed in line with DG 365: *Soakaway design* (BRE, 2016).

The design method for sizing the soakaway should be based on the equation of volumes:

$$I - O = S$$

where I is the inflow from the impermeable area drained to the soakaway (m^3), O is the outflow infiltrating into the soil during rainfall (m^3) and S is the required storage in the soakaway to balance the temporary inflow and outflow (m^3).

Inflow to the soakaway:

$$I = A \times R$$

where A is the impermeable area drained to the soakaway (m^2) and R is the total rainfall in a design storm (m).

Outflow to the soakaway:

$$O = a_{s50} \times f \times D$$

where a_{s50} is the internal surface area of the soakaway to 50% effective depth (excluding the base area, which is assumed to clog with fine particles and become ineffective in the long term) (m^2), f is the soil infiltration rate determined by use of a trial pit at the site of the soakaway (m/s) and D is the storm duration.

Stormwater attenuation should be based on a 1:5-year storm event with a duration of 24 hours, as per local authority guidelines.

It is very important to note that surface water soakaways should not normally be constructed closer than 5 metres from the building foundations. Site investigations must be undertaken to determine the soil conditions, as this is crucial in the viability and sizing of the soakaway

14 Fuel gas design consideration

The design of the LPG system should adhere to the safety of human beings, whether or not they are installing the system. Modern technology should be adopted to detect any failure in the system through a high-tech gas control panel, linked to the fire control panel, to ensure the maintenance of a safe environment.

Design of liquid gas distribution systems should be in accordance with the certified and applicable specifications and engineering standards, and must be submitted with the drawings for approval of building permits transactions.

Only Civil Defence approved material, control and reading devices should be used.

It is preferable to place the natural gas tank within the limits of the plot, whether below ground level or above, and with safe distances compatible with national safety requirements and the requirements of Saudi Civil Defence.

The central tank should be supplied with means of security, protection and control in accordance with standards specified by Saudi Civil Defence and/or NFPA 58 (NFPA, 2020), and should comply with all standards and special requirements for public safety adopted by Saudi local requirements.

Execution of domestic gas networks should be performed only by specialised companies registered with the Saudi Authority.

First stage (with two safety devices):

- LPG entry to building: 350 mbar (as per NFPA 58, Section 3.2.13 C, (NFPA, 2020), this can be up to 1.4 bar)
- natural gas entry to building: 75 mbar.

Second stage (with one safety device)

- LPG inside kitchen: 37 mbar
- natural gas inside kitchen: 21 mbar.

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